



Course Code and Title: ITE104 – Information Management

Lesson 5-6: Week 5-6

Topic: End User Computing

Introduction

End-user computing (EUC) refers to computer systems and platforms that help non-programmers create applications. However, there is a lot more to EUC and its related technology, virtual desktop infrastructure (VDI), which essentially hosts desktop environments on a central server. It is considered a form of desktop virtualization.

Lesson Presentation:

End User Computing

End-user computing (EUC) is a term that refers to the technologies that IT professionals use to deploy, manage and secure the devices, applications and data that workers require to perform their jobs.

The major components of EUC are physical desktop computing, virtual desktop computing and mobile computing, each of which involves several different technologies.

Types of EUC

End-user computing encompasses a wide variety of user-facing resources, including:

1. desktop and notebook computers;
2. desktop operating systems and applications;
3. smartphones, tablets, wearables and other mobile devices;
4. mobile, web and cloud applications; and
5. virtual desktops and applications.

EUC also covers the technologies that IT professionals use to provide access to these resources, such as:

- Windows management and security tools;



management and mobile application management;

- desktop and application virtualization platforms and management tools; and
- enterprise file sync-and-share services.

End-user computing services

Traditionally, IT managed the different components of end-user computing separately. As the consumerization of IT and the bring your own device (BYOD) trend gained steam, however, more organizations realized the need to provide access to corporate applications and data across multiple device types.

In an attempt to simplify this process, vendors began offering products and services designed to work across multiple areas of EUC. Examples of these products and services include:

- tools that provide monitoring and management of both physical and virtual desktops and applications;
- app refactoring, which uses virtualization to create mobile-friendly versions of Windows and web apps;
- unified endpoint management (UEM), which allows IT to apply and enforce mobile device management policy on Windows 10 PCs; and
- workspace suites, which aim to provide centralized consoles where end users can access all of their required applications and data, and IT can securely manage that access.

Advantages and disadvantages

End-user computing benefits organizations by securely enabling a mobile, distributed workforce. Its major disadvantage, despite all the work that has gone into unified



management, is its complexity.

Most organizations have not yet migrated to Windows 10, which means they can't take advantage of UEM, and they have to use separate products to manage PCs and mobile devices. Workspace suites do not yet provide complete integration between all of the disparate products they bundle together. And trying to run an application on an operating system or device that it wasn't built for can result in problems around compatibility and user experience.

Office Automation

- Office automation refers to the use of computer and software to digitally create, collect, store, manipulate, and relay office information needed for accomplishing basic tasks and goals.
- The use of computer systems to do a variety of office operations, such as word processing, accounting, and e-mail.
- Raw data storage, electronic transfer, and the management of electronic business information comprises the basic activities of an office automation system.
- Office automation helps in optimizing or automating existing office procedures
- The backbone of office automation is a LAN, which allows users to transmit data, mail and even voice across the network. All office functions, including dictation, typing, filing, copying, fax, Telex, microfilm and records management, telephone and telephone switchboard operations, fall into this category.

Benefits of an Office Automation System

Implementing an office automation system offers organizations a broad range of benefits. These benefits include:

- Benefits of an Office Automation System
- Improved accuracy



- Reduced costs
- Reduced time and resources
- Data storage and management
- Data insights and more informed decisions
- Business process improvement

Improved accuracy - Humans make errors, properly implemented automated systems do not. Human errors are not only inefficient in that they must be corrected and lead to productivity delays, but they can be costly. For instance, adding too many digits when paying an employee or vendor. Serious mistakes can lead to security and compliance issues, potentially fines and penalties. An office automation system limits human intervention in the transfer of data, which minimizes the occurrence of errors.

Reduced costs - By automating complex business processes, organizations don't need to invest as much into hiring for those tasks. As a result, operational costs are lower, while productivity and profit margins are significantly higher.

Reduced time and resources - Through automation, organizations can accomplish more with less. By eliminating tedious and time-consuming processes, employees can spend their time on more high value tasks. For example, returning to the popular paperless benefit, with office automation software employees do not need to spend as much time collecting important documentation, entering that information, or filing away voluminous paper copies. The system does it for them. In fact, 78% of leaders anticipate automation could free up 3 hours a day of an employee's time.

Data storage and management - Office automation systems simplify data storage while giving organizations the ability to monitor and control data through an electronic document management system. Common features include things like task management and reminder systems, as well as easy access to information by key stakeholders.



Data insights and more informed decisions - Office automation systems give organizations access to large data sets, reports, and analytics. Access to data enables more informed decisions. Moreover, by analyzing data and key performance indicators, organizations can implement improvements to their processes to remove bottlenecks and other inefficiencies.

Through business process improvement, organizations optimize performance, improve the quality of their products or services, and ensure a higher level of compliance. The ability to improve business processes is what separates office automation systems from piecemeal automation technology.

Key features of an office automation system

There are countless applications and software programs that promise improved efficiency through automation. As we discussed above, however, achieving organizational efficiency through an office automation system requires more than downloading a few apps. Rather, you need a comprehensive office automation solution that offers the following features:

Process modeling and workflow design - An important benefit of an office automation system is the ability to design and improve workflows. The solution that you choose should give you the ability to create detailed workflows in a matter of minutes. Leading solutions offer an intuitive design and include features like a drag and drop process modeler.

Mobile compatibility - In the age of the stay-at-home economy and the COVID-19 pandemic, mobility is more important than ever. Employees must be able to perform their functions from anywhere. A cloud-based solution is always secure and always accessible.

Integration - Task management is an important feature of office automation solutions. For processes to run smoothly, employees must know what to do and



when to do it. Your office automation system should allow you to create and view pending tasks and deadlines, as well as redistribute tasks as needed.

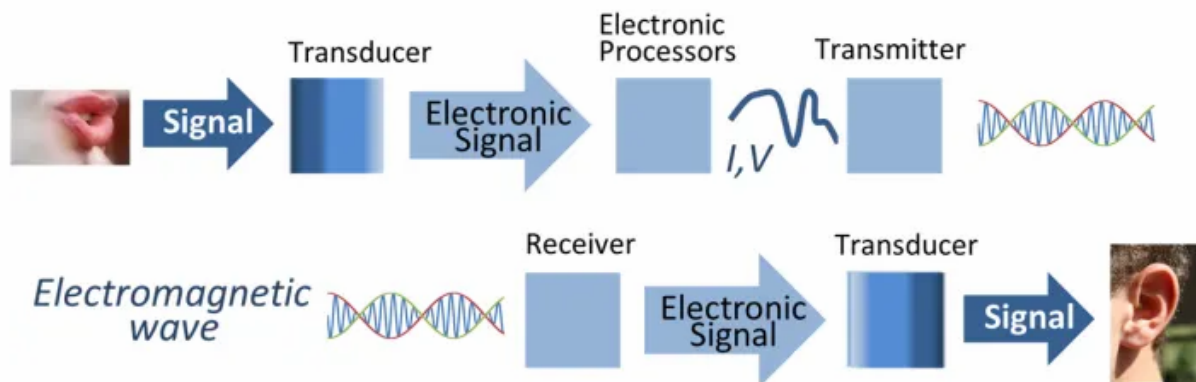
Access control and security - To protect the security of your systems, you need to be able to set access privileges throughout the organization. Cloud-based solutions also offer advanced security features to protect data from being compromised.

Communication - An important feature of any automation system is the ability to communicate seamlessly. Stakeholders should have all the information they need to perform each task, as well as the ability to reach out to others for further assistance. Your office automation system should offer features like a form builder. Form builders can be used to capture data, display data from other systems, and even design approval screens for managers to make decisions.

Reporting and analytics - To evaluate and improve your office automation system, you need to have access to data and key performance indicators. Leading office automation systems have built-in reporting and analytics capabilities.

Element of Electronic Communication System

Electronic communications is the transmission, reception, and processing of information between two or more locations with the use of electronic circuits. The basic components of electronic communications system are the transmitter, communications channel or medium, receiver, and noise. Analog signals (such human voice) or digital signals (binary data) are inputted to the system, processed in the electronic circuits for transmission, and then decoded by the receiver. The system is said to be reliable and effective only when errors are minimized in the process.



A sample diagram as to how signal is being processed in electronic communications.

Transmitter - Transmitter is defined as a collection of one or more electronic devices or circuits that converts the original source information - also called as baseband signal - to a form suitable for transmission. It is a part of the system where the sender inputs or encodes the information. Examples of transmitters are our mobile phones and AM radio transmitters.

Functions of a Radio Transmitter

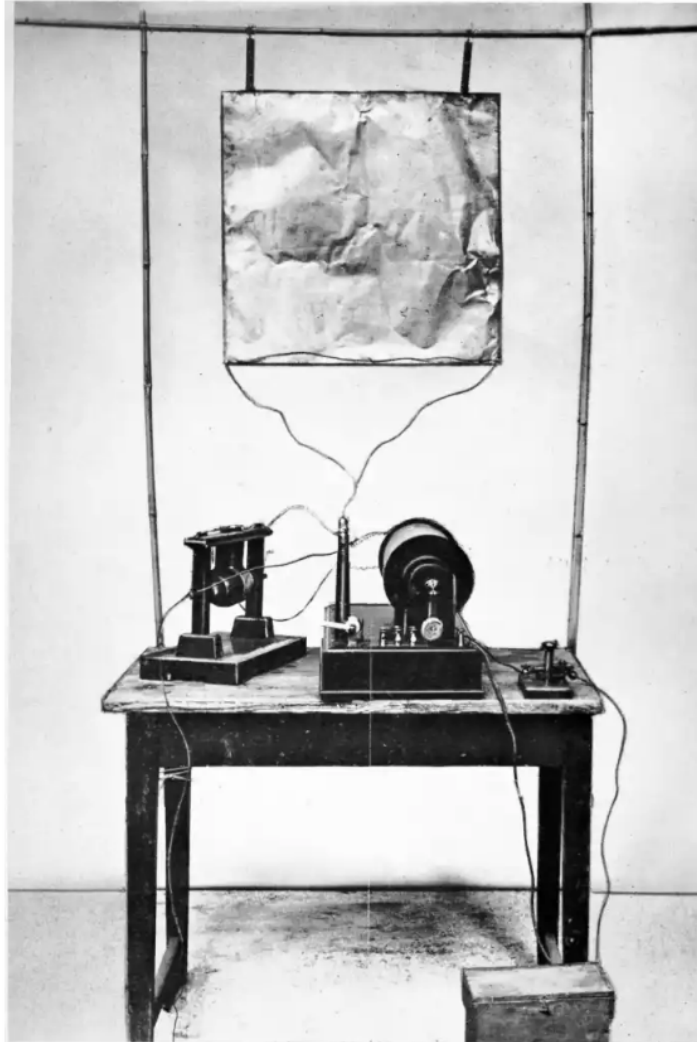
A radio transmitter is an electronic unit that accepts the information signal to be transmitted and converts it into a radio frequency (RF) signal capable of being transmitted over long distances.

A transmitter must:

- Generate a signal of desired frequency
- Provide some form of modulation that allows the information signal to modify a signal of higher frequency, also known as the carrier signal. Amplitude Modulation (AM) and Frequency Modulation are commonly used in broadcasting.
- Provide power amplification in order to ensure that the signal level is high, in



such a way that it will carry over the desired distance for which the signal is to be sent.



Communications channel

The channel is the medium by which the electronic signal is sent or propagated from one place to another. It provides a means of transporting signals between a transmitter and a receiver. The channel can be as simple as copper wire or as complex as the optical fiber and satellite systems. Signals can also be propagated through radio waves or free space depending upon the type of modulation and frequency being used.



Although channel provides a way for communication, it also attenuates the signal that carries the message. All types of media are capable of degrading the signals, resulting to weaker signals, and appear to have smaller amplitudes. Amplifiers are included both in the transmitter and receiver to compensate this problem.



Receiver

The receiver is a collection of electronic devices and circuits that accepts the transmitted signals from the transmission medium and then converts those signals back to their original form which is understandable by humans. Good examples of receivers are radio devices and television sets.



Primary requirements for any communications receiver:

Selectivity - The ability of a receiver to select a signal of a desired frequency while rejecting those on closely adjacent frequencies. With good selectivity, the receiver can select the desired signal and eliminate all other RF signals. Tuned circuits or LC circuits are used to obtain selectivity.

Sensitivity - It is the ability of a receiver to pick up weak signals. Sensitivity is directly related to receiver's gain. By the way, gain is the factor by which an input signal is multiplied to produce the output. This can be increased by having a series of amplification. Better sensitivity is attained with higher gain.

This is how noise degrades the quality of a signal



System noise

- Noise is any unwanted electrical signals that interfere with the information signal. It is a random electrical energy that enters the system via the medium. However, noise can also be generated in the receiver which causes some errors in demodulation process.
- Noise may come from different sources such as the atmosphere, particularly lightning and thunderstorms. It may also come from the outer space where the sun and billions of stars emit radiation which interferes with the signal.
- Noise is inevitable, it is always present in a system. It can never be avoided, but can always be minimized with the use of sophisticated techniques and electronic circuits.

Electronic Conferencing

Electronic conferencing is a widespread method used in today's businesses, as well as in education and social networking websites. The purpose of electronic



conferencing is for individuals to engage in various dialogues without being in the same room. Participants can also send and receive files, as well as share documents onto a conferencing discussion board.

Web Conferencing

One form of e-conferencing, video or web conferencing, is used so everyone who may not be able to sit in a room can physically be together in the virtual realm. Video conferencing combines audio and visual elements to bring people together for a discussion or meeting. To achieve video conferencing, a computer, webcam, microphone and Internet must be used to share information, provide interaction among each participant, and provide people with the ability to see each other throughout the conferencing.

Conferencing Forums

With electronic conferencing, several conferences are held online in a chat forum. From instant messaging to a chat discussion on a website or online classroom, conferencing forums can bring a variety of information and opinions to people who may be in the same city--or even in different countries.

Software

People use various types of e-conferencing software during a conferencing session. For social chat forums, MSN, Yahoo Messenger and Skype are the most common software used for casual conversations among friends and associates. Even social networking sites MySpace and Facebook have chat forums for members to communicate with each other. For business purposes, software including iLinc, Picture Talk and even Microsoft Office provides private chat rooms for business partners and associates to discuss meetings and plans for a company.

Moderator

In all discussion forums, a moderator is present to help in discussion as well as be a



referee whenever an individual or discussion is out of context. The moderator doesn't express her own views or give advice to members inside a forum; she's there to solve issues or provide general instructions on how to make the discussion run smoothly. Other duties of a moderator include welcoming people to a discussion room, introducing topics to participants, displaying effective use of the discussion forum, and solving conflicts and issues among participants in the forum.

Electronic Meeting System

An electronic meeting system (EMS) is software intended to provoke problem-solving and decision-making within a group. Standard features of electronic meeting systems include electronic brainstorming (generally in an anonymous format), parallel processing, discussion tools, and voting. The unique features of an EMS overcome the limitations of traditional face-to-face meetings such as lack of participation, criticism, and domination of the proceedings by only a few members.

Understanding Electronic Meeting Systems

Electronic meeting systems (EMS) are distinct from web or video conferencing meeting systems, though both have some features in common and complement each other in the modern workplace. An EMS offers many advantages to organizations and users, such as a reduction in travel costs, increased participation, due to the anonymity the system offers, fewer disruptions compared to meeting in a busy office, and better participant availability.

These advantages more than offset the major drawback of such systems, which is a lack of personal contact and human interaction. There are clear advantages to being able to read body language and make eye contact when communicating with other people. An EMS typically blocks or at least limits this type of interaction. One other drawback to an EMS is the risk of technological or electronic interruptions, should the digital connectivity become unstable or even give out.



KEY TAKEAWAYS

- An electronic meeting system (EMS) is a software platform that provides a digital meeting space for group collaboration, problem-solving, and providing anonymous feedback, among other features.
- Electronic meeting systems help companies by cutting down on travel costs and by making participation easier because everything's anonymous and accessible from wherever the worker is located.
- An EMS can never replace the personal contact that a more traditional meeting might facilitate; an EMS is also vulnerable if a facility is having connectivity issues, which would make communicating online difficult.

Example of Electronic Meeting Systems:

An EMS can be designed for limited or specific purposes such as training and sales presentations. An example of potential use for an EMS would be when a company plans to launch a new product and wants to test the potential appeal among employees who have the background and knowledge to be objective about the proposed offering. Using an electronic meeting system, participants can share opinions and criticism free of potential conflicts and groupthink that might otherwise affect responses.



Summary:

End-User Computing (EUC) strategies that had only focused on device and internal cost optimization activities will be replaced and use workspace analytics will be adopted to predict end-user needs and deliver a better experience. EUC will be much more flexible and unified with condition access policy that enables to have control on how authorized users can access cloud apps.

Even Google and Microsoft are partnering with End-user Computing firms to create a solution that can turn on remote applications on any devices and widely improving user experience and making it easier for IT departments.

Data will be centralized and to be more secured than ever before. IT departments can manage workspace anywhere from one point, and the End-user device is optimized for faster and easier to use.

Reference:

<https://searchmobilecomputing.techtarget.com/definition/end-user-computing-platform-EUC-platform>

End-user computing (EUC)

<https://www.processmaker.com/blog/implementing-an-office-automation-system/>

Implementing an Office Automation System

<https://turbofuture.com/industrial/Elements-of-Electronic-Communications-System>

Elements of Electronic Communications System

<https://www.investopedia.com/terms/e/electronic-meeting-system.asp>

Electronic Meeting System



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